

Package ‘breaktest’

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Title Set of unit root and cointegration tests

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Depends R (>= 3.5.0)

Encoding UTF-8

Imports doSNOW, foreach, stringr, Rfast

Description This package contains a set of different unit root and
cointegration tests in the presence of structural breaks in the data.

License GPL (>= 2)

URL <https://github.com/d9d6ka/breaktest>

BugReports <https://github.com/d9d6ka/breaktest/issues>

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.ic.values	<i>Information criteria</i>
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Description

Information criteria

Usage

.ic.values(resids, extra, modification = FALSE, alpha = 0, y = NULL)

Arguments

resids	Input residuals needed for estimating the values of information criteria.
extra	Number of extra parameters needed for estimating the punishment term.
modification	Whether the unit-root test modifacaton is needed. See Ng and Perron (2001) for further information.
alpha	The coefficient α of y_{t-1} in ADF model. Needed only for criterion modification purposes.
y	The vector of y_{t-1} in ADF model. Needed only for criterion modification purposes.

Details

Calculating the value of the following informational criteria:

- Akaike,
- Schwarz (Bayesian),
- Hannan-Quinn,
- Liu et al.

Value

The list of information criteria values.

References

Ng, Serena, and Pierre Perron. "Lag Length Selection and the Construction of Unit Root Tests with Good Size and Power." *Econometrica* 69, no. 6 (2001): 1519–54. <https://doi.org/10.1111/1468-0262.00256>.

ADF.test

A simple implementation of ADF test

Description

A function for ADF test with the ability to select the number of lags. Lags are selected by informational criteria which can be modified as in Ng and Perron (2001) and Cavaliere et al. (2015).

Usage

```
ADF.test(
  y,
  const = TRUE,
  trend = FALSE,
  max.lag = 0,
  criterion = NULL,
  modified.criterion = FALSE,
  rescale.criterion = FALSE
)
```

Arguments

<code>y</code>	A time series of interest.
<code>const, trend</code>	Whether a constant and trend are to be included.
<code>max.lag</code>	Maximum lag number.
<code>criterion</code>	A criterion used to select number of lags. If lag selection is not needed keep this NULL.
<code>modified.criterion</code>	Whether the unit-root test modification is needed.
<code>rescale.criterion</code>	Whether the rescaling informational criterion is needed. Designed to cope with heteroscedasticity in residuals.

Details

Due to the Frisch-Waugh-Lovell theorem we first detrend `y` and then apply the test to the detrended series.

Value

A list containing:

- `y`,
- `const`,
- `trend`,

- residuals,
- coefficient estimates,
- t-statistic value,
- critical value,
- Number of lags,
- indicator of stationarity.

References

Cavaliere, Giuseppe, Peter C. B. Phillips, Stephan Smeekes, and A. M. Robert Taylor. “Lag Length Selection for Unit Root Tests in the Presence of Nonstationary Volatility.” *Econometric Reviews* 34, no. 4 (April 21, 2015): 512–36. <https://doi.org/10.1080/07474938.2013.808065>.

Ng, Serena, and Pierre Perron. “Lag Length Selection and the Construction of Unit Root Tests with Good Size and Power.” *Econometrica* 69, no. 6 (2001): 1519–54. <https://doi.org/10.1111/1468-0262.00256>.

ADF.test.S

Detrending bootstrap test by Smeekes (2013)

Description

This bootstrap test is based on the recursive detrending procedure of Taylor (2002). The main idea is to apply the standard ADF test to the series with nuisance parameters eliminated.

Usage

```
ADF.test.S(
  y,
  const = TRUE,
  trend = FALSE,
  c = 0,
  gamma = 0,
  trim = 0.15,
  max.lag = 0,
  criterion = NULL,
  modified.criterion = FALSE,
  iter = 999
)
```

Arguments

y	A time series of interest.
const, trend	Whether the constant and trend are to be included.
c	A filtration parameter used to construct an autocorrelation coefficient.
gamma	Detrending type selection parameter. If 0 the OLS detrending is applied, if 1 the GLS detrending is applied, otherwise the autocorrelation coefficient is calculated as $1 + c^\gamma T^{-\gamma}$.
trim	A trimming parameter.

max.lag	The maximum lag for inner ADF testing.
criterion	A criterion used to select number of lags. If lag selection is not needed keep this NULL.
modified.criterion	Whether the unit-root test modification is needed.
iter	The number of bootstrap iterations.

Details

Critical values are calculated via a bootstrapping using MacKinnon-like regressions. For each number of observations and each number of variables obtained were 1999 values of test statistics. After that 1st, 2.5-th, 5-th, 10-th, and 97.5-th percentiles were calculated and saved along with the corresponding number of observations. This step was repeated 5 times to cope with possible biases. After that MacKinnon-like regressions were estimated.

References

- Taylor, A. M. Robert. "Regression-Based Unit Root Tests With Recursive Mean Adjustment for Seasonal and Nonseasonal Time Series." *Journal of Business & Economic Statistics* 20, no. 2 (April 2002): 269–81. <https://doi.org/10.1198/073500102317352001>.
- MacKinnon, James G. "Critical Values for Cointegration Tests." Working Paper. Economics Department, Queen's University, January 2010. <https://ideas.repec.org/p/qed/wpaper/1227.html>.
- Smeekes, Stephan. "Detrending Bootstrap Unit Root Tests." *Econometric Reviews* 32, no. 8 (July 2013): 869–91. <https://doi.org/10.1080/07474938.2012.690693>.
- Elliott, Graham, Thomas J. Rothenberg, and James H. Stock. "Efficient Tests for an Autoregressive Unit Root." *Econometrica* 64, no. 4 (1996): 813–36. <https://doi.org/10.2307/2171846>.

coint.conf.sets

Confidence sets for the break date in cointegrating regressions

Description

This procedure is to construct a confidence set for the change point in cointegrating regressions.

Usage

```
coint.conf.sets(
  y,
  trend = FALSE,
  zb = NULL,
  zf = NULL,
  z.lead = NULL,
  z.lag = NULL,
  conf.level = 0.9,
  trim = 0.05,
  criterion = "bic"
)
```

Arguments

<code>y</code>	A time series of interest.
<code>trend</code>	Whether the trend is to be included.
<code>zb</code>	I(1) regressors with break.
<code>zf</code>	I(1) regressors without break.
<code>z.lead, z.lag</code>	Number of leads and lags of <code>z</code> regressors. If any is NULL then both are estimated using informational criterion.
<code>conf.level</code>	Confidence level to obtain appropriate critical values.
<code>trim</code>	The trimming parameter to find the lower and upper bounds of possible break date.
<code>criterion</code>	A criterion for lead and lag number estimation.

Value

A list of confidence sets.

References

Kurozumi, Eiji, and Anton Skrobotov. "Confidence Sets for the Break Date in Cointegrating Regressions." *Oxford Bulletin of Economics and Statistics* 80, no. 3 (2018): 514–35. <https://doi.org/10.1111/obes.12223>.

coint.test.GH

Gregory-Hansen test for the absense of cointegration

Description

Gregory and Hansen (1996) test for the null hypothesis of no cointegration under a possible structural break at the unknown moment of time.

The authors proposed ADF- and Z-type tests, slightly modified to allow the presence of a possible regime shift. Three type of shifts are allowed:

- a shift in the constant,
- a shift in the constand with the trend included,
- and a shift in the constant and the cointegrating vector.

Critical values are calculated via the adopted MacKinnon procedure of estimating the model for the response surface.

Usage

```
coint.test.GH(
  ...,
  shift = "level",
  trim = 0.15,
  max.lag = 10,
  criterion = "aic",
  add.cvals = TRUE
)
```

Arguments

...	Variables of interest.
shift	Expected break type.
trim	The trimming parameter to calculate break moment bounds.
max.lag	The maximum number of lags for the internal ADF testing.
criterion	The criterion for lag selection.
add.cvals	Whether critical values are to be returned. This argument is needed to suppress the calculation of critical values during the precalculation of tables needed for the p-values estimating.

Value

An object of type cointGH. It's a list of

- shift: shift type,
- Za: MZ_α statistic and c.v.,
- Zt: MZ_t statistic and c.v.,
- ADF: ADF statistic and c.v..

References

MacKinnon, James G. "Critical Values for Cointegration Tests." Working Paper. Working Paper. Economics Department, Queen's University, January 2010. <https://ideas.repec.org/p/qed/wpaper/1227.html>.

Gregory, Allan W., and Bruce E. Hansen. "Residual-Based Tests for Cointegration in Models with Regime Shifts." *Journal of Econometrics* 70, no. 1 (January 1, 1996): 99–126. [https://doi.org/10.1016/0304-4076\(99\)41685-7](https://doi.org/10.1016/0304-4076(99)41685-7).

coint.test.PR	<i>A set of residual based tests for cointegration</i>
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Description

A set of residual based tests for cointegration

Usage

```
coint.test.PR(y, x, deter, min.lag = 0)
```

Arguments

y, x	Variables of interest. x can be a matrix of several variables.
deter	A value equal to • 1: quasi-demeaned y and x, • 2: quasi-detrended y and x, • 3: quasi-demeaned y and quasi-detrended x.
min.lag	A minimum number of lags to be used in the tests.

Details

The code provided is the original GAUSS code by Perron and Rodríguez ported to R.

Value

A list of:

- 7x1-matrix of test statistics values,
- estimated number of lags.

References

Perron, Pierre, and Gabriel Rodríguez. “Residuals-based Tests for Cointegration with Generalized Least-squares Detrended Data.” *The Econometrics Journal* 19, no. 1 (February 1, 2016): 84–111. <https://doi.org/10.1111/ectj.12056>.

KP	<i>Kejrival-Perron procedure of breaks number detection</i>
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Description

Kejrival-Perron procedure of breaks number detection

Usage

```
KP(y, const = FALSE, breaks = 1, criterion = "aic", trim = 0.15)
```

Arguments

y	An input series of interest.
const	Whether the break in constant is allowed.
breaks	Number of breaks.
criterion	Needed information criterion: aic, bic, hq or lwz.
trim	A trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

The estimated optimal break point.

References

Kejriwal, Mohitosh, and Pierre Perron. “A Sequential Procedure to Determine the Number of Breaks in Trend with an Integrated or Stationary Noise Component: Determination of Number of Breaks in Trend.” *Journal of Time Series Analysis* 31, no. 5 (September 2010): 305–28. <https://doi.org/10.1111/j.1467-9892.2010.00666.x>.

KPSS.1br	<i>KPSS-test with known structural break</i>
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Description

Computes the cointegration test with one known structural break.

Usage

```
KPSS.1br(y, x, model, bp, weakly.exog = TRUE, ll.init)
```

Arguments

y	A time series of interest.
x	A matrix of explanatory stochastic regressors.
model	A scalar equal to • 1: for model An, • 2: for model A, • 3: for model B, • 4: for model C, • 5: for model D, • 6: for model E.
bp	A position of the break point.
weakly.exog	Exogeneity of the stochastic regressors • TRUE: if the regressors are weakly exogenous, • FALSE: if the regressors are not weakly exogenous (DOLS is used in this case).
ll.init	A scalar, defines the initial number of leads and lags for DOLS.

Details

The code provided is the original GAUSS code by Carrion-i-Silvestre and Sansó ported to R.
See Carrion-i-Silvestre and Sansó (2006) for further details.

Value

A list of:

- beta: DOLS estimates of the coefficients,
- tests: SC test (coinKPSS-test),
- resid: Residuals of the model,
- t.beta: Individual significance t-statistics,
- break_point: Break point.

References

Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. “Testing the Null of Cointegration with Structural Breaks.” *Oxford Bulletin of Economics and Statistics* 68, no. 5 (October 2006): 623–46. <https://doi.org/10.1111/j.1468-0084.2006.00180.x>.

KPSS.1br.unknown	<i>KPSS-test of cointegration</i>
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Description

Procedure for testing the null of cointegration in the possible presence of structural breaks.

Usage

```
KPSS.1br.unknown(y, x, model, weakly.exog, ll.init)
```

Arguments

y	A time series of interest.
x	A matrix of explanatory stochastic regressors.
model	A scalar equal to • 1: for model An, • 2: for model A, • 3: for model B, • 4: for model C, • 5: for model D, • 6: for model E.
weakly.exog	Exogeneity of the stochastic regressors • TRUE: if the regressors are weakly exogenous, • FALSE: if the regressors are not weakly exogenous (DOLS is used in this case).
ll.init	Scalar, defines the initial number of leads and lags for DOLS.

Details

Computes the cointegration test with one unknown structural break where the break point is estimated either minimizing the value of the statistic or the sum of the squared residuals. The estimation of the cointegrating relationship bases on DOLS.

The code provided is the original GAUSS code ported to R.

See Carrion-i-Silvestre and Sansó (2006) for further details.

Value

(2x2)-matrix, where the first rows gives the value of the min(SC) test and the estimated break point; the second row gives the value of the SC statistic, where the break point is estimated as min(SSR).

References

Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. “Testing the Null of Cointegration with Structural Breaks.” *Oxford Bulletin of Economics and Statistics* 68, no. 5 (October 2006): 623–46. <https://doi.org/10.1111/j.1468-0084.2006.00180.x>.

KPSS.2br

KPSS-test with 2 known structural breaks

Description

Procedure to compute the KPSS test with two structural breaks

Usage

```
KPSS.2br(y, model, bp, max.lag, kernel)
```

Arguments

y	A time series of interest.
model	A scalar equal to • 1: for the AA (without trend) model, • 2: for the AA (with trend) model, • 3: for the BB model, • 4: for the CC model, • 5: for the AC-CA model.
max.lag	A scalar, with the maximum order of the parametric correction. The final order of the parametric correction is selected using the BIC information criterion.
kernel	Kernel for calculating long-run variance • bartlett: for Bartlett kernel, • quadratic: for Quadratic Spectral kernel, • NULL for the Kurozumi's proposal, using Bartlett kernel.
break.point	Positions for the first and second structural breaks (relative to the origin which is 1).

Details

The break points are known.

The code provided is the original GAUSS code by Carrion-i-Silvestre and Sansó ported to R.

See Carrion-i-Silvestre and Sansó (2007) for further details.

Value

A list of:

- beta: DOLS estimates of the coefficients,
- tests: SC test (coinKPSS-test),
- resid: Residuals of the model,
- t.beta: *t*-statistics for beta,
- break_point: Break points.

References

Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. "The KPSS Test with Two Structural Breaks." *Spanish Economic Review* 9, no. 2 (May 16, 2007): 105–27. <https://doi.org/10.1007/s10108-006-9017-8>.

 KPSS.2br.unknown

KPSS-test with 2 unknown structural breaks

Description

Procedure to compute the KPSS test with two structural breaks

Usage

```
KPSS.2br.unknown(y, model, max.lag = 0, kernel = "bartlett")
```

Arguments

<code>y</code>	A time series of interest.
<code>model</code>	A scalar equal to • 1: for the AA (without trend) model, • 2: for the AA (with trend) model, • 3: for the BB model, • 4: for the CC model, • 5: for the AC-CA model.
<code>max.lag</code>	A scalar, with the maximum order of the parametric correction. The final order of the parametric correction is selected using the BIC information criterion.
<code>kernel</code>	Kernel for calculating long-run variance • <code>bartlett</code>: for Bartlett kernel, • <code>quadratic</code>: for Quadratic Spectral kernel, • <code>NULL</code> for the Kurozumi's proposal, using Bartlett kernel.

Details

The break points are known

The code provided is the original GAUSS code ported to R.

See Carrion-i-Silvestre and Sansó (2007) for further details.

Value

Value of test statistic.

References

Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. "Testing the Null of Cointegration with Structural Breaks." *Oxford Bulletin of Economics and Statistics* 68, no. 5 (October 2006): 623–46. <https://doi.org/10.1111/j.1468-0084.2006.00180.x>.

 KPSS.HLT

Unit root testing procedure under a single structural break.

Description

Unit root testing procedure under a single structural break.

Usage

```
KPSS.HLT(y, const = FALSE, trim = 0.15)
```

Arguments

y	A time series of interest.
const	Whether a constant should be included.
trim	The trimming parameter to find the lower and upper bounds of possible break dates.

Value

The value of test statistic.

References

Harvey, David I., Stephen J. Leybourne, and A. M. Robert Taylor. "Unit Root Testing under a Local Break in Trend." *Journal of Econometrics* 167, no. 1 (2012): 140–67.

 KPSS.mlt

KPSS-test with multiple known structural breaks

Description

Procedure to compute the KPSS test with multiple known structural breaks

Usage

```
KPSS.mlt(
  y,
  x,
  model,
  bp,
  const = FALSE,
  trend = FALSE,
  weakly.exog = TRUE,
  lags.init,
  leads.init,
  max.lag,
  kernel,
  criterion = "bic"
)
```

Arguments

<code>y</code>	A time series of interest.
<code>x</code>	A matrix of explanatory stochastic regressors.
<code>model</code>	A scalar or vector of • 1: for the break in const, • 2: for the break in trend, • 3: for the break in const and trend.
<code>bp</code>	Array of structural breaks.
<code>const, trend</code>	Whether a constant or trend should be included.
<code>weakly.exog</code>	Boolean where we specify whether the stochastic regressors are exogenous or not • TRUE: if the regressors are weakly exogenous, • FALSE: if the regressors are not weakly exogenous (DOLS is used in this case).
<code>lags.init, leads.init</code>	Scalars defining the initial number of lags and leads for DOLS.
<code>max.lag</code>	scalar, with the maximum order of the parametric correction. The final order of the parametric correction is selected using the BIC information criterion.
<code>kernel</code>	Kernel for calculating long-run variance • <code>bartlett</code>: for Bartlett kernel, • <code>quadratic</code>: for Quadratic Spectral kernel, • <code>NULL</code> for the Kurozumi's proposal, using Bartlett kernel.
<code>criterion</code>	Information criterion for DOLS lags and leads selection: <code>aic</code> , <code>bic</code> , <code>hq</code> , or <code>lwz</code> .

Details

The code provided is based on the original code by Carrion-i-Silvestre and Sansó.

Value

A list of

- `beta`: DOLS estimates of the coefficients,
- `tests`: SC test (coinKPSS-test),
- `resid`: Residuals of the model,
- `t.beta`: *t*-statistics for `beta`,
- `DOLS.lags`: The estimated number of lags and leads in DOLS,
- `break_point`: Break points.

References

- Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. "Testing the Null of Cointegration with Structural Breaks." *Oxford Bulletin of Economics and Statistics* 68, no. 5 (October 2006): 623–46. <https://doi.org/10.1111/j.1468-0084.2006.00180.x>.
- Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. "The KPSS Test with Two Structural Breaks." *Spanish Economic Review* 9, no. 2 (May 16, 2007): 105–27. <https://doi.org/10.1007/s10108-006-9017-8>.

KPSS.mlt.bootstrap	<i>KPSS-test with multiple unknown structural breaks</i>
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Description

Procedure to compute the KPSS test with multiple unknown structural breaks

Usage

```
KPSS.mlt.bootstrap(
  y,
  x,
  model,
  bp,
  const = FALSE,
  trend = FALSE,
  weakly.exog = TRUE,
  lags.init,
  leads.init,
  max.lag,
  kernel,
  iter = 9999,
  bootstrap = "sample",
  criterion = "bic"
)
```

Arguments

y	A time series of interest.
x	A matrix of explanatory stochastic regressors.
model	A scalar or vector of • 1: for the break in const, • 2: for the break in trend, • 3: for the break in const and trend.
const	Include constant if TRUE .
trend	Include trend if TRUE .
weakly.exog	Boolean where we specify whether the stochastic regressors are exogenous or not • TRUE: if the regressors are weakly exogenous, • FALSE: if the regressors are not weakly exogenous (DOLS is used in this case).
lags.init, leads.init	Scalars defining the initial number of lags and leads for DOLS.
max.lag	scalar, with the maximum order of the parametric correction. The final order of the parametric correction is selected using the BIC information criterion.
kernel	Kernel for calculating long-run variance • bartlett: for Bartlett kernel,

	- quadratic: for Quadratic Spectral kernel, - NULL for the Kurozumi's proposal, using Bartlett kernel.
iter	Number of bootstrap iterations.
bootstrap	Type of bootstrapping: "sample": sampling from residuals with replacement, "Cavalieri-Taylor": multiplying residuals by $N(0, 1)$-distributed variable, "Rademacher": multiplying residuals by Rademacher-distributed variable.
criterion	Information criterion for DOLS lags and leads selection: aic, bic or lwz.
break.point	Array of structural breaks.

Value

A list of:

- test: The value of KPSS test statistic,
- p.value: The estimates p-value,
- bootstrapped: Bootstrapped auxiliary statistics.

MDF.1br

MDF procedure for a single unknown break.

Description

MDF procedure for a single unknown break.

Usage

```
MDF.1br(y, const = FALSE, trend = FALSE, trim = 0.15)
```

Arguments

y	A time series of interest.
const	Whether the constant term should be included.
trend	Whether the trend term should be included.
trim	Trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

A list of sublists each containing

- The value of statistic: $MDF - GLS$, $MDF - OLS$,
- The asymptotic critical values. UR values are included as well.

MDF.CHLT

*MDF test for a single break and possible heteroscedasticity***Description**

MDF test for a single break and possible heteroscedasticity

Usage

```
MDF.CHLT(y, max.lag = 10, trim = 0.15, iter = 499)
```

Arguments

<code>y</code>	A time series of interest.
<code>max.lag</code>	The maximum possible lag.
<code>trim</code>	Trimming parameter for lag selection
<code>iter</code>	Number of bootstrap iterations.

Details

The code provided is the original GAUSS code by Cavaliere et al. ported to R.

Value

An object of type `mdfCHLT`. It's a list of four sublists each containing:

- The value of MZ_α , MSB , MZ_t , or ADF ,
- The asymptotic c.v.,
- The bootstrapped c.v.

References

Cavaliere, Giuseppe, David I. Harvey, Stephen J. Leybourne, and A.M. Robert Taylor. "Testing for Unit Roots in the Presence of a Possible Break in Trend and Nonstationary Volatility." *Econometric Theory* 27, no. 5 (October 2011): 957–91. <https://doi.org/10.1017/S0266466610000605>.

MDF.mlt

*MDF procedure for multiple unknown breaks.***Description**

MDF procedure for multiple unknown breaks.

Usage

```
MDF.mlt(y, const = FALSE, breaks = 1, breaks.star = 1, trim = 0.15, ZA = FALSE)
```

Arguments

y	A time series of interest.
const	Whether the constant term should be included.
breaks	Number of breaks.
breaks.star	Number of breaks got from the Kejriwal-Perron procedure.
trim	Trimming value for a possible break date bounds.
ZA	Whether ZA variant should be used.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

A list of sublists each containing

- The value of statistic: $MDF - GLS$, $MDF - OLS$,
- The asymptotic critical values. UR values are included as well.

PY.sequential	<i>Sequential Perron-Yabu (2009) statistic for breaks at unknown date.</i>
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Description

Sequential Perron-Yabu (2009) statistic for breaks at unknown date.

Usage

```
PY.sequential(
  y,
  const = FALSE,
  breaks = 1,
  criterion = "aic",
  trim = 0.15,
  max.lag = 1
)
```

Arguments

y	A time series of interest.
const	Allowing the break in constant.
breaks	A number of breaks.
criterion	Needed information criterion: aic, bic, hq or lwz.
trim	A trimming value for a possible break date bounds.
max.lag	The maximum possible lag in the model.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

An estimated Wald statistic.

References

Kejriwal, Mohitosh, and Pierre Perron. “A Sequential Procedure to Determine the Number of Breaks in Trend with an Integrated or Stationary Noise Component: Determination of Number of Breaks in Trend.” *Journal of Time Series Analysis* 31, no. 5 (September 2010): 305–28. <https://doi.org/10.1111/j.1467-9892.2010.00666.x>.

PY.single	<i>Perron-Yabu (2009) statistic for break at unknown date.</i>
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Description

Perron-Yabu (2009) statistic for break at unknown date.

Usage

```
PY.single(
  y,
  const = FALSE,
  trend = FALSE,
  criterion = "aic",
  trim = 0.15,
  max.lag
)
```

Arguments

y	A time series of interest.
const, trend	Allowing the break in constant or trend.
criterion	Needed information criterion: aic, bic, hq or lwz.
trim	A trimming parameter to determine the lower and upper bounds for a possible break point.
max.lag	The maximum possible lag in the model.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

A list of the estimated Wald statistic as well as its c.v.

References

Perron, Pierre, and Tomoyoshi Yabu. “Testing for Shifts in Trend With an Integrated or Stationary Noise Component.” *Journal of Business & Economic Statistics* 27, no. 3 (July 2009): 369–96. <https://doi.org/10.1198/jbes.2009.07268>.

`robust.tests.multiple` *A wrapping function around [KP](#) and [MDF.mlt](#).*

Description

A wrapping function around [KP](#) and [MDF.mlt](#).

Usage

```
robust.tests.multiple(
  y,
  const = FALSE,
  season = FALSE,
  breaks = 2,
  trim = 0.15
)
```

Arguments

<code>y</code>	A series of interest.
<code>const</code>	Whether the constant term should be included.
<code>season</code>	Whether the seasonal adjustment is needed.
<code>breaks</code>	Number of breaks.
<code>trim</code>	Trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

`robust.tests.single` *A wrapping function around [MDF.lbr](#).*

Description

A wrapping function around [MDF.lbr](#).

Usage

```
robust.tests.single(
  y,
  const = FALSE,
  trend = FALSE,
  season = FALSE,
  trim = 0.15
)
```

Arguments

y	A time series of interest.
const, trend	Whether the constant term and trend should be included.
season	Whether the seasonal adjustment is needed.
trim	Trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

SADF.bootstrap.test *Supremum ADF tests with wild bootstrap.*

Description

SADF.bootstrap.test is a wild bootstrapping procedure for estimating critical and p -values for [SADF.test](#).

GSADF.bootstrap.test is the same procedure but for GSADF.test.

Usage

```
SADF.bootstrap.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 999,
  seed = round(10^4 * sd(y))
)

GSADF.bootstrap.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 4 * 200,
  seed = round(10^4 * sd(y))
)
```

Arguments

y	A time series of interest.
trim	A trimming parameter to determine the lower and upper bounds for a possible break point.
const	Whether the constant needs to be included.
alpha	The significance level of interest.
iter	The number of iterations.
seed	The seed parameter for the random number generator.

Details

Refactored original code by Kurozumi et al.

Value

An object of type `sadf`. It's a list of:

- `y`,
- `trim`,
- `const`,
- `alpha`,
- `iter`,
- `seed`,
- vector of t -values,
- the value of the corresponding test statistic,
- series of bootstrapped test statistics,
- bootstrapped critical values,
- p -value.

References

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." *Journal of Financial Econometrics*, April 23, 2022. <https://doi.org/10.1093/jfinec/nbac004>.

SADF.test	<i>Supremum ADF tests</i>
-----------	---------------------------

Description

SADF.test is a test statistic equal to the minimum value of ADF.test for subsamples starting at $t = 1$.

GSADF.test is a generalized version of SADF.test. Subsamples are allowed to start at any point between 1 and $T(1 - trim)$.

Usage

```
SADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  add.p.value = TRUE
)

GSADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  add.p.value = TRUE
)
```

Arguments

y	A time series of interest.
trim	A trimming parameter to determine the lower and upper bounds for a possible break point.
const	Whether the constant needs to be included.
add.p.value	Whether the p-value is to be returned. This argument is needed to suppress the calculation of p-values during the precalculation of tables needed for the p-values estimating.

Details

Refactored original code by Kurozumi et al.

Value

An object of type `sadf`. It's a list of:

- y,
- trim,
- const,
- vector of t -values,
- the value of the corresponding test statistic,
- p -value if it was asked for.

References

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." *Journal of Financial Econometrics*, April 23, 2022. <https://doi.org/10.1093/jjfinec/nbac004>.

sb.GSADF.test

Sign-based SADF test

Description

Sign-based SADF test

Usage

```
sb.GSADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 999,
  urs = TRUE,
  seed = round(10^4 * sd(y))
)
```

Arguments

y	A time series of interest.
trim	A trimming parameter to determine the lower and upper bounds for a possible break point.
const	Whether the constant needs to be included.
alpha	Needed level of significance.
iter	Number of bootstrapping iterations.
urs	Use union of rejections strategy if TRUE.
seed	The seed parameter for the random number generator.

Details

Refactored original code by Kurozumi et al.

References

Harvey, David I., Stephen J. Leybourne, and Yang Zu. “Sign-Based Unit Root Tests for Explosive Financial Bubbles in the Presence of Deterministically Time-Varying Volatility.” *Econometric Theory* 36, no. 1 (February 2020): 122–69. <https://doi.org/10.1017/S0266466619000057>.

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. “Time-Transformed Test for Bubbles under Non-Stationary Volatility.” *Journal of Financial Econometrics*, April 23, 2022. <https://doi.org/10.1093/jjfinec/nbac004>.

STADF.test

Supremum ADF tests with time transformation

Description

See [SADF.test](#). Tests with time transformation are the modified versions of the ordinary SADF and GSADF tests using Nadaraya-Watson residuals and reindexing procedure by Cavaliere-Taylor (2008).

Usage

```
STADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = FALSE,
  omega.est = TRUE,
  truncated = TRUE,
  is.reindex = TRUE,
  ksi.input = "auto",
  hc = 1,
  pc = 1,
  add.p.value = TRUE
)

GSTADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = FALSE,
```



```

    omega.est = TRUE,
    truncated = TRUE,
    is.reindex = TRUE,
    ksi.input = "auto",
    hc = 1,
    pc = 1,
    add.p.value = TRUE
)

```

Arguments

<code>y</code>	A time series of interest.
<code>trim</code>	A trimming parameter to determine the lower and upper bounds for a possible break point.
<code>const</code>	Whether the constant needs to be included.
<code>omega.est</code>	Whether the variance of Nadaraya-Watson residuals should be used.
<code>truncated</code>	Whether the truncation of Nadaraya-Watson residuals is needed.
<code>is.reindex</code>	Whether the Cavaliere and Taylor (2008) time transformation is needed.
<code>ksi.input</code>	The value of the truncation parameter. Can be either auto or the explicit numerical value. In the former case the numeric value is estimated.
<code>hc</code>	The scaling parameter for Nadaraya-Watson bandwidth.
<code>pc</code>	The scaling parameter for the estimated truncation parameter value.
<code>add.p.value</code>	Whether the p-value is to be returned. This argument is needed to suppress the calculation of p-values during the precalculation of tables needed for the p-values estimating.

Details

Refactored original code by Kurozumi et al.

Value

An object of type `sadf`. It's a list of:

- `y`,
- `N`: Number of observations,
- `trim`,
- `const`,
- `omega.est`,
- `truncated`,
- `is.reindex`,
- `new.index`: the vector of new indices,
- `ksi.input`,
- `hc`,
- `h.est`,
- `u.hat`,
- `pc`,

- `w.sq`,
- `t.values`: vector of t -values,
- the value of the corresponding test statistic,
- `u.hat.truncated`: truncated residuals if truncation was asked for,
- `ksi, sigma`: estimated values of the truncation parameter and resulting s.e. if `ksi.input` equals `auto`,
- `eta.hat`: the values of reindexing function if reindexing was asked for,
- p -value if it was asked for.

References

Cavaliere, Giuseppe, and A. M. Robert Taylor. "Time-Transformed Unit Root Tests for Models with Non-Stationary Volatility." *Journal of Time Series Analysis* 29, no. 2 (March 2008): 300–330. <https://doi.org/10.1111/j.1467-9892.2007.00557.x>.

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." *Journal of Financial Econometrics*, April 23, 2022. <https://doi.org/10.1093/jfinec/nbac004>.

VECM.test

Test of the cointegration rank with a possible break in trend

Description

This procedure is aimed on the problem of testing for the cointegration rank of a vector autoregressive process in the case where a trend break may potentially be present in the data.

The test is based on estimating the quasi log likelihood for two situations, with break, and without it. The one with the smallest value is considered to be the result.

Usage

```
VECM.test(y, r, max.lag, trim = 0.15)
```

Arguments

<code>y</code>	A matrix of n VAR variables.
<code>r</code>	The cointegration rank tested against the alternative of n .
<code>max.lag</code>	The maximum number of lags.
<code>trim</code>	The trimming parameter to find the lower and upper bounds of possible break dates.

Details

The code provided is the original GAUSS code by Harris et al. ported to R.

Value

A list of:

- the indicator of the rejection of null.
- the estimated break point.
- the estimated lag number.

References

Harris, David, Stephen J. Leybourne, and A. M. Robert Taylor. “Tests of the Co-Integration Rank in VAR Models in the Presence of a Possible Break in Trend at an Unknown Point.” *Journal of Econometrics, Innovations in Multiple Time Series Analysis*, 192, no. 2 (June 1, 2016): 451–67. <https://doi.org/10.1016/j.jeconom.2016.02.010>.

weighted.SADF.test	<i>Weighted supremum ADF test</i>
--------------------	-----------------------------------

Description

Weighted supremum ADF test

Usage

```
weighted.SADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 4 * 200,
  urs = TRUE,
  seed = round(10^4 * sd(y))
)

weighted.GSADF.test(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 4 * 200,
  urs = TRUE,
  seed = round(10^4 * sd(y))
)
```

Arguments

y	A time series of interest.
trim	The trimming parameter to find the lower and upper bounds of possible break dates.
const	Whether the constant needs to be included.
alpha	A significance level of interest.
iter	Nnumber of iterations.
urs	Use union of rejections strategy.
seed	A seed parameter for the random number generator.

Details

Refactored original code by Kurozumi et al.

Value

An object of type `sadf`. It's a list of:

- `y`,
- `trim`,
- `const`,
- `alpha`,
- `iter`,
- `urs`,
- `seed`,
- `sigma.sq`: the estimated variance,
- `BZ.values`: a series of BZ-statistic,
- `supBZ.value`: the maximum of `supBZ.values`,
- `supBZ.bootstrap.values`: bootstrapped supremum BZ values,
- `supBZ.cr.value`: supremum BZ α critical value,
- `p.value`,
- `is.explosive`: 1 if `supBZ.value` is greater than `supBZ.cr.value`.

if `urs` is TRUE the following items are also included:

- vector of t -values,
- the value of the SADF test statistic,
- `SADF.bootstrap.values`: bootstrapped SADF values,
- `U.value`: union test statistic value,
- `U.bootstrap.values`: bootstrapped series of `U.value`,
- `U.cr.value`: critical value of `U.value`.

References

- Harvey, David I., Stephen J. Leybourne, and Yang Zu. "Testing Explosive Bubbles with Time-Varying Volatility." *Econometric Reviews* 38, no. 10 (November 26, 2019): 1131–51. <https://doi.org/10.1080/07474938.2019.1644444>
- Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." *Journal of Financial Econometrics*, April 23, 2022. <https://doi.org/10.1093/jfinec/nbac004>.

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